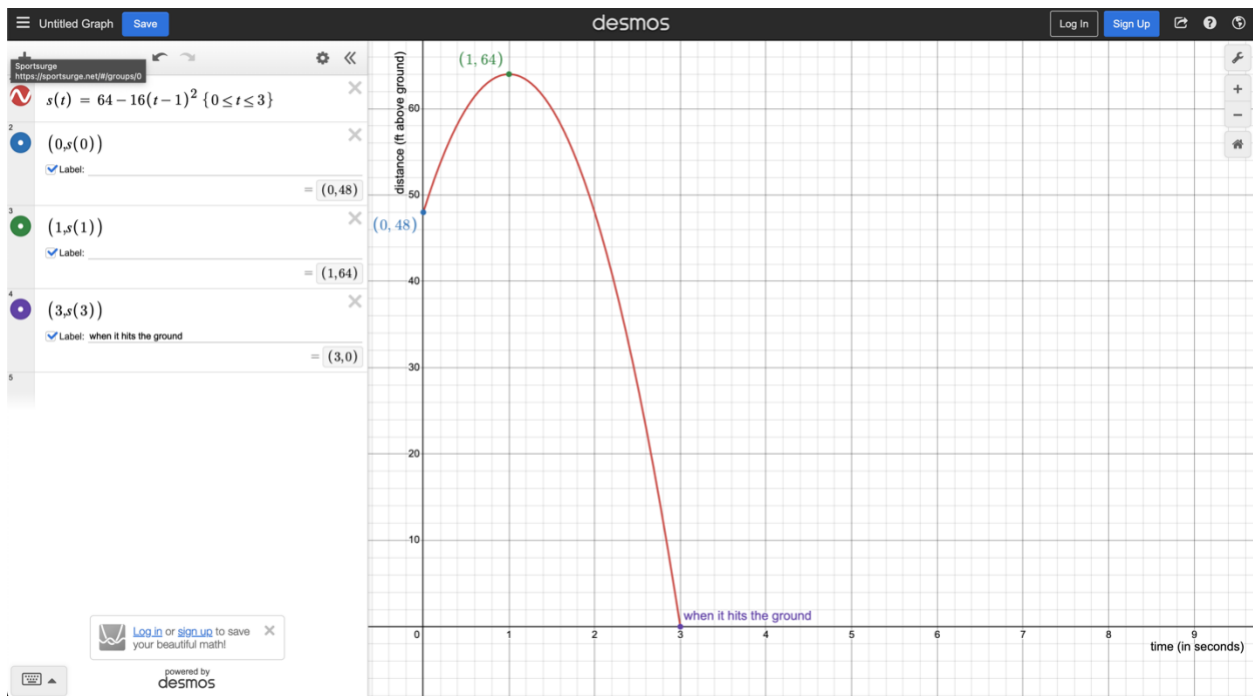


Preview Activity 1.1.1

a.



b.

During time interval of $0 < t < 1$ the ball is traveling upwards after it has been tossed up. During time interval of $1 < t < 3$ the ball is traveling downwards due to the pull of gravity. At $t=1$ the ball is changing directions from traveling in a positive direction to traveling in a negative direction. That is where our slope (and our derivative) will = 0

c.

Handwritten calculation of average velocity on the interval $[0.5, 1]$:

$$AV_{[0.5, 1]} = \frac{s(1) - s(0.5)}{1 - 0.5} \Rightarrow \frac{64 - 60}{.5} = \frac{4}{.5} = 8$$

This is avg velocity on the time interval of 0.5 to 1

Alternative Average Velocity Formula Video

a.

$$13.36 + 13.6h - 16h^2$$

b.

They're not the same. If you have $s(0.2 + h)$ it means you are plugging in the entire expression into the function. Whereas $s(0.2) + h$ would just be $+ h$ to whatever the output of $s(0.2)$ would be.

Average Velocity Applet

1. 80mph

2. 60mph

3. -40mph

4. 64mph

5. Between 2 and 2.5 hours